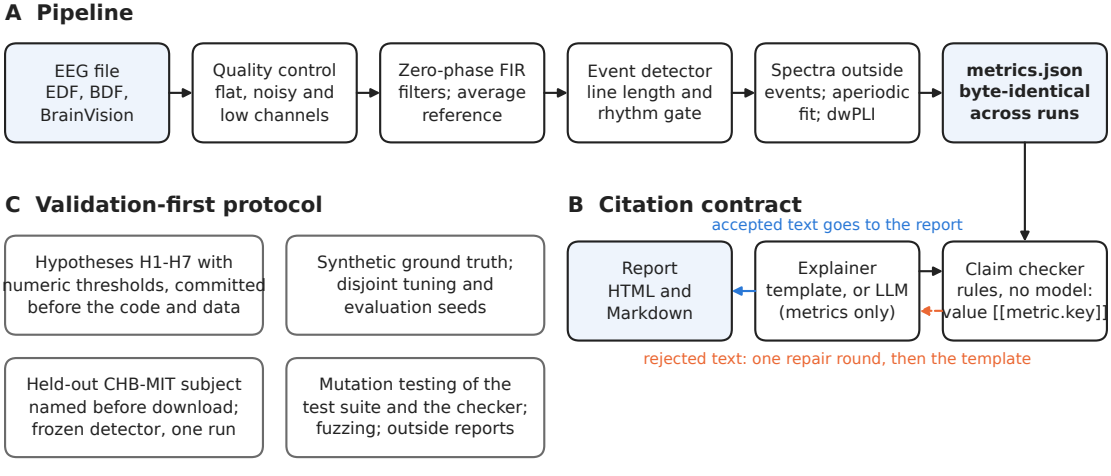
A Pipeline
EEG file EDF, BDF, BrainVision
Quality control flat, noisy and low channels
Zero-phase FIR filters; average reference
Event detector line length and rhythm gate
Spectra outside events; aperiodic fit; dwPLI
metrics.json byte-identical across runs
B Citation contract
accepted text goes to the report
Report HTML and Markdown
Explainer template, or LLM (metrics only)
Claim checker rules, no model: value [[metric.key]]
rejected text: one repair round, then the template
C Validation-first protocol
Hypotheses H1-H7 with numeric thresholds, committed before the code and data
Synthetic ground truth; disjoint tuning and evaluation seeds
Held-out CHB-MIT subject named before download; frozen detector, one run
Mutation testing of the test suite and the checker; fuzzing; outside reports